THREE EQUIVALENT CONDITIONS FOR A FINITELY PRESENTED CLOSED SUBSCHEME OF THE SOURCE OF AN ALMOST BLOWUP SQUARE TO CONTAIN THE RELEVANT PIECES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EV8 (sources/stacks-flat/pinned-original-source.tex line 11027).

Allowed standard prerequisites (editorial).

- Almost blowup squares and the description of the ideal sheaf of the blowup as sections supported on E. Same chapter, flat.tex, Equation labelled equation-almost-blow-up-square (source line 10957), invoked in the statement. Supplies the square and its defining property, which is what makes condition (1) imply condition (2).
- Isomorphism of the complement of Z in X with the complement of E in X'. Same chapter, flat.tex, the structural properties of almost blowup squares used at proof lines 11052-11054. Supplies X" cap (X' setminus E) is X setminus Z is X' setminus E, which is the step from (3) back to (1).

Source reference. Original equation 0EV6: sources/stacks-flat/pinned-original-source.tex, source lines 10957–10957. Same-chapter pointer into the author environment equation at source line 10957; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Consider an almost blow up square (0EV6). Let $W \to X'$ be a closed immersion of finite presentation. The following are equivalent*

(1) *$X' \setminus E$ is scheme theoretically contained in W,*
(2) *the blowup X'' of X in Z is scheme theoretically contained in W,*

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(3) *the diagram*

$$\begin{array}{ccc} E \cap W & \longrightarrow & W \\ \downarrow & & \downarrow \\ Z & \longrightarrow & X \end{array}$$

is an almost blow up square. Here $E \cap W$ is the scheme theoretic intersection.

Proof. Assume (1). Then the surjection $\mathcal{O}_{X'} \to \mathcal{O}_W$ is an isomorphism over the open $X' \subset E$. Since the ideal sheaf of $X'' \subset X'$ is the sections of $\mathcal{O}_{X'}$ supported on E (by our definition of almost blow up squares) we conclude (2) is true. If (2) is true, then (3) holds. If (3) holds, then (1) holds because $X'' \cap (X' \setminus E)$ is isomorphic to $X \setminus Z$ which in turn is isomorphic to $X' \setminus E$. $\square$